DWDM

Class Test

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Q1 (a) Data Mining and KDD process

1. Define Data Mining:

Data mining is the process of discovering useful patterns, relationships, and trends from large amounts of data.

1. classification: Assigns data to predefined categories. Example: classifying emails as spam or not spam.
2. clustering: Groups similar data items together. Example: grouping customers based on shopping habits

2. KDD process:

Knowledge discovering in Databases is the process of extracting useful knowledge from large datasets.

1. Data selection: select relevant data from different sources
2. Data preprocessing: Remove errors, handle missing values, and clean the data.
3. Data Transformation: convert data into a suitable format for mining.
4. Data Mining: Apply algorithms to discover patterns and relationships.
5. Pattern Evaluation: Identify useful and meaningful pattern.
6. Knowledge Presentation: Present the discovered knowledge using reports, graphs or visualization.

3. Role of data preprocessing in

- Improves data quality and accuracy.
- Handles missing and noisy data.
- Removes duplicate and inconsistent records
- Improves the efficiency of data mining algorithms
- Helps produce reliable result.

(b) Data cleaning and Data Quality Issues

1. Data cleaning:

Data cleaning is the process of identifying and correcting or removing inaccurate, incomplete, inconsistent, and duplicate data from a dataset.

1. Missing values: some fields in a dataset contain no value.
2. Noisy data: Data contains random errors or unusual values.

2. Methods to handle missing values.

- Ignore the record: Remove records containing missing values when appropriate.
- Fill manually: Enter the correct value after checking the original source
- Replace with a constant: Use a value with the mean or median.
- Mode Imputation: Replace missing categorical values with the most frequent value.

3. Methods to handle noisy data.

- Binning: group values and smooth them using bin means, median or boundaries
- Regression: use a Mathematical model to estimate values and reduce random errors.
- Outlier detection: Identify unusually high or low values and investigate them.

Q2(a) Data Integration.

Data integration is the process of combining data from multiple sources into a single, unified dataset

1. Redundancy: the same data may appear in multiple sources
2. Inconsistency: the same data may have different values or formates in different sources.

Handling reduncy and inconsistency:

- Redundancy: Identify duplicate attributes or records and remove unnecessary duplicates. correlation analysis can help identify redundant attributes.
- Inconsistency: Detect conflicting values and resolve them using standard formats, data rules, and verification from reliable sources

(b) Covariance Analysis

Covariance analysis is a statistical method used to measure how two variables change together.

1. Positive covariance: Both variables tend to increase or decrease together.
2. Negative covariance: When one variable increase, the other tends to decrease.

Example: Product price and demand.

Q3. a) covariance of two stocks

Stock A (X)	Stock B (Y)
2	5
3	8
5	10
4	11
6	14

$\bar{X} = \frac{2+3+5+4+6}{5} = 4$ $\quad \bar{Y} = \frac{5+8+10+11+14}{5} = 9.6$

Covariance:

$$cov(X,Y) = \frac{\Sigma (x_i - \bar{x})(y_i - \bar{y})}{n}$$

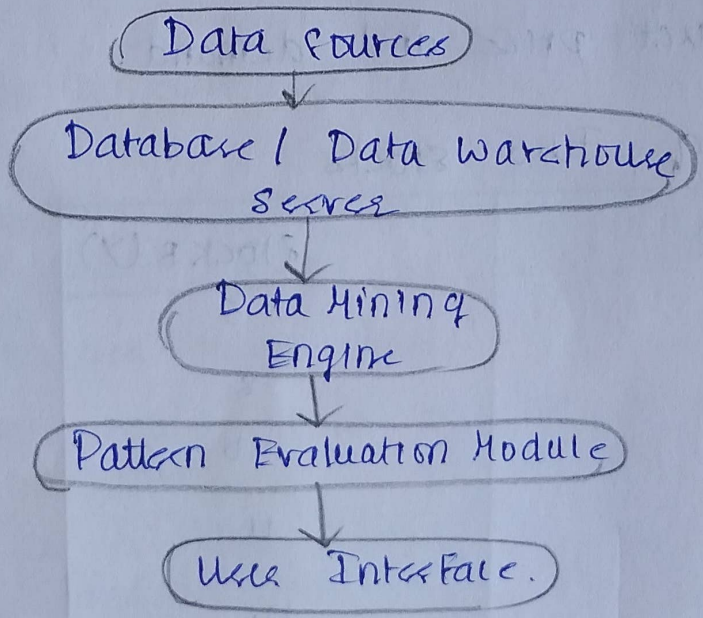
X	Y	(X-4)	(Y-9.6)	Product
2	5	-2	-4.6	9.2
3	8	-1	-1.6	1.6
5	10	1	0.4	0.4
4	11	0	1.4	0
6	14	2	4.4	8.8

$$cov(X,Y) = \frac{20}{5} = 4$$

∴ covariance = 4 (positive) thus the stock tend to rise or fall together showing a positive relationship rather than independence.

b) Architecture of a Typical Data Mining system

A data mining system discovers useful patterns and knowledge from large amounts of data.



Components and their Roles:

1. Data sources: Provide raw data from databases, files, and other sources.
2. Database: Retrieves and manages the required data
3. Data Mining Engine: Applies algorithms such as classification, clustering, and association rule mining.
4. Pattern Evaluation Module: Identifies useful and interesting patterns and filters irrelevant results
5. Knowledge Base: store domain knowledge, rules and Metadata to guide the mining process.
6. User interface: Allows users to specify queries and view the results through reports and visualization.

24: (a) Find covariance

$X = \{4, 5, 6, 7, 9\}$ $Y = \{8, 3, 7, 5, 6\}$

$\bar{X} = \frac{4+5+6+7+9}{5} = \frac{31}{5} = 6.2$ $\bar{Y} = \frac{8+3+7+5+6}{5} = \frac{29}{5} = 5.8$

$$cov(X, Y) = \frac{\sum (x_i - \bar{X})(y_i - \bar{Y})}{n}$$

X	Y	$(X-6.2)$	$(Y-5.8)$	Product
4	8	-2.2	2.2	-4.84
5	3	-1.2	-2.8	3.36
6	7	-0.2	1.2	-0.24
7	5	0.8	-0.8	-0.64
9	6	2.8	0.2	0.56

$$cov(x, y) = \frac{+1.8}{5} = 0.36$$

(b) Major steps in Data preprocessing

Data preprocessing is the process of converting raw data into a clean and suitable form for data mining

1. Data cleaning: Handles missing values, noisy data, and duplicate records
2. Data Integration: Combine data from multiple sources and resolves inconsistencies
3. Data Transformation: Normalizes, aggregate or converts data into suitable formats.
4. Data reduction: Reduces the data size while preserving useful information.
5. Data Discretization: converts continuous numerical values into intervals or categories

Preprocessing pipeline:

1. Identify missing values, outliers, and format errors
2. Standardize formats and units
3. Handle missing values using mean, median or mode
4. Detect and correct or remove verified outliers.
5. Transforms and validates the cleaned data.

Q3 (a) Equal Frequency Binning

Bin 1: 13, 15, 16, 16, 19, 20, 20, 21, 22

$$\text{Mean} = \frac{162}{9} = 18$$

Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35

$$\text{Mean} = \frac{253}{9} = 28.11$$

Bin 3: 35, 35, 35, 36, 40, 45, 46, 52, 70

$$\text{Mean} = 394/9 = 43.78$$

b) Data Discretization

Data Discretization converts continuous numerical values into a limited number of intervals or categories.

Importance:

- Simplifies data
- Reduce noise
- Makes patterns easier to understand
- Helps some data mining algorithms

Comparision

Equal-width binning	Equal-Frequency binning
Bins have equal ranges	Bins contain approximately equal number of value
Bin counts may differ	Bin widths may differ
Sensitive to extreme values	Useful for unevenly distributed data

Impact: Equal width binning gives consistent ranges, while equal-Frequency bining balances the number of observation in each bin.